

Grade 7 Accelerated
Unit 1
Lesson 2 Homework

Name _____
Date _____
Period _____

Consider the repeating decimal $0.\overline{23}$ for questions 1-4.

1. If $x = 0.\overline{23}$, complete each equation.

$$x = 0.\overline{23}$$

$$10x = \underline{\hspace{2cm}}$$

$$100x = \underline{\hspace{2cm}}$$

2. When subtracted, which two values would result in the repeating portion of the decimal being eliminated?

3. Use the two relationships represented in question 1 to complete the steps to find the fraction equivalent of $0.\overline{23}$.

$$\underline{\hspace{2cm}} - \underline{\hspace{2cm}} = \underline{\hspace{2cm}} - \underline{\hspace{2cm}}$$

$$\underline{\hspace{2cm}}x = \underline{\hspace{2cm}}$$

$$x = \frac{\boxed{\hspace{1cm}}}{\boxed{\hspace{1cm}}}$$

4. Write the fraction and mixed number that is equivalent to $8.\overline{23}$.

A trail mix recipe calls for the following ratios: $26.\overline{6}\%$ raisins, $\frac{3}{8}$ peanuts, 0.09 chocolate chips, and the rest is made up of chex mix.

5. What fraction of the recipe is made up of chocolate chips?
6. What fraction of the recipe is made up of raisins?
7. What is the decimal equivalent of the peanuts?
8. Approximately what percent is made up of Chex Mix? Round your answer to the nearest whole percent.

For questions 9-10, complete the table below.

	Decimal	Percent	Fraction (or Mixed Number)
9.	$4.\overline{65}$		
10.		$6.2\overline{4}\%$	